

From spill events to catchment-scale prediction

Environmental drivers and predictability of combined sewer overflow duration behaviour across England

Project focus. A national analysis linking event-level CSO monitoring records to wastewater treatment works (WWTW) catchments, fitted long-duration tail parameters, environmental characteristics, one-variable statistical analysis and out-of-sample multivariable prediction.

Author

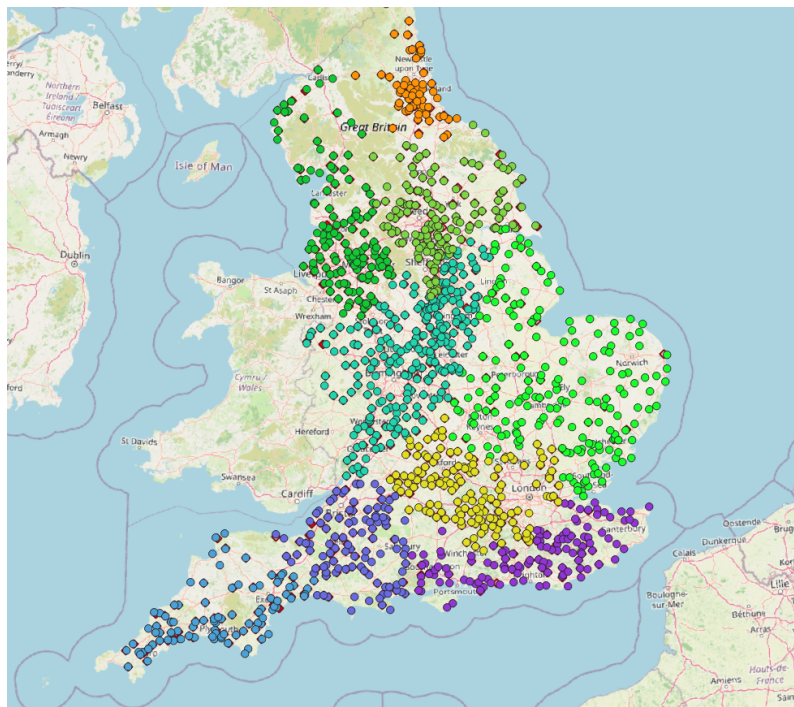
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Research setting

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This report follows the logic of a technical research paper: problem, data, parameterisation, spatial linkage, one-variable evidence, multivariable predictive modelling, synthesis and limitations. The visual language is intentionally restrained so the scientific argument carries the report.

Abstract

Combined sewer overflows (CSOs) are usually summarised by how often they spill or how long they spill in total. Those summaries do not fully describe the persistence of the longest events. This project constructs a national event-level workflow that begins with water-company Event Duration Monitoring records and ends with four related but distinct outcomes at WWTW-associated sensors: the fitted long-duration tail-shape parameter β , the fitted inverse-timescale parameter λ , mean spill duration, and annualised spill frequency. Events lasting more than 240 minutes define the fitted long-duration tail. Sensor records are linked to WWTW sewersheds, treatment capacity and a common external predictor framework containing rainfall, hydrogeology, land cover, population, infrastructure-age proxy, terrain and catchment scale. The analysis deliberately progresses from unadjusted one-variable associations to a leakage-safe multivariable supervised-regression programme evaluated on unseen WWTWs. The central scientific question is not simply which variables correlate with CSO behaviour, but how much of the spatial variation can be explained and predicted from measured external catchment characteristics. The final predictive programme finds modest rather than decisive out-of-sample skill, which is itself informative: static catchment descriptors contain structure, but important event-scale hydraulic and temporal information remains unobserved.

Executive summary

THE QUESTION

Can the environmental and operational characteristics of a WWTW catchment explain why its associated CSO sensor has a particular long-duration spill distribution, and can those characteristics predict behaviour at an unseen WWTW?

THE ANALYTICAL STORY

The project first created a common national event dataset, then quantified the long-duration tail using β and λ . Those sensor-level outcomes were made interpretable spatially by linking sensors to WWTW sewersheds. External catchment variables were then extracted at the same spatial unit. One-variable regressions first identified the direction and scale of isolated associations. The final machine-learning stage then placed the external variables together in multivariable scikit-learn regression models and asked whether that combined information could generalise to unseen WWTWs.

Table 1: Current final predictive scorecard. RMSE values are on the logged model scale and are only comparable within a target.

Target	Selected model	Test RMSE	Dummy RMSE	Test R^2	Interpretation
β	OLS, interpretable static	0.653	0.687	0.073	4.9% lower RMSE; modest signal, bootstrap improvement still includes zero.
λ	OLS, interpretable static	3.855	4.056	0.095	4.9% lower RMSE; weak and inconclusive because improvement uncertainty includes zero.
Mean duration	Ridge, static + annual rainfall	1.631	1.675	0.052	2.6% lower RMSE; modest signal, with substantial extra value from company context.
Annualised frequency	Ridge, static + annual rainfall	1.341	1.442	0.133	7.0% lower RMSE; clearest baseline improvement of the four new models.

THE CENTRAL CONCLUSION

Measured spatial context is informative but not sufficient for precise site-level prediction. Population and hydrogeology carry signal for the fitted tail parameters, rainfall contributes to duration, and intergranular-flow geology plus annual dry-spell information contribute to annualised frequency. Yet the learning curves are already close to a plateau. The next major gain is therefore more likely to come from richer hydraulic and time-resolved process information than from fitting a more complicated algorithm to the same static variables.

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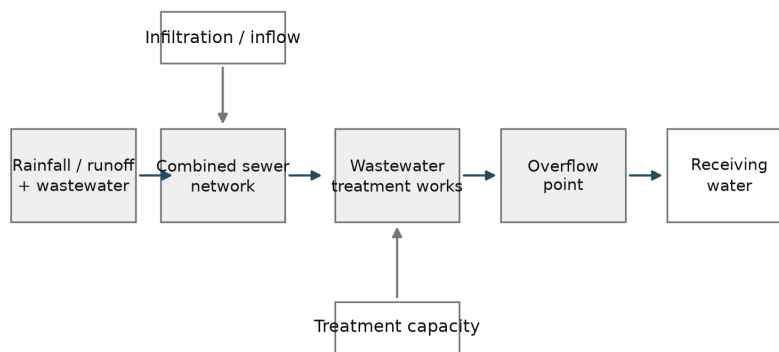
1 Introduction

1.1 Why long-duration CSO behaviour matters

A combined sewer overflow is a pressure-relief mechanism in a sewer network. During sufficiently high hydraulic loading, diluted sewage and stormwater can discharge to the environment rather than continuing through the network and treatment process. Public reporting often focuses on event counts or annual duration. Those quantities are important, but two sites with similar counts can still have very different distributions of event duration. One may experience many relatively contained events, while another has a much more persistent upper tail.

This project therefore asks a different question: *what controls the shape and timescale of the longest spills, and how predictable are those characteristics from the external environment of the wastewater catchment?*

A combined sewer links weather, wastewater treatment and receiving waters



Schematic only - not a hydraulic network model

Figure 1: Conceptual overview of combined sewer overflow behaviour and the long-duration spill process analysed in this study.

1.2 The four outcomes

The final analysis deliberately separates four quantities that answer different physical questions.

Table 2: Four outcomes used throughout the report.

Outcome	What it measures	Interpretation
Tail shape β	Shape of the duration distribution conditional on an event lasting > 240 min	Lower β means a heavier, more persistent long-duration tail; higher β means a faster-decaying tail.
Inverse timescale λ	Fitted scale component in $S(t) = \exp[-(\lambda t)^\beta]$	At fixed β , higher λ means a shorter characteristic time $1/\lambda$. It is not conventional Weibull η .
Mean spill duration	Observed mean duration of valid events	Direct site-level duration burden, expressed in minutes before transformation.
Annualised spill frequency	Valid event count divided by documented monitoring years	Exposure-adjusted frequency; preferable to raw lifetime count when monitoring periods differ.

1.3 Research questions

The report is organised around five linked questions:

1. What does the national event-level CSO duration distribution look like, particularly above four hours?
2. Can stable sensor-specific β and λ values be estimated from those long-duration events?
3. Can those sensor outcomes be linked credibly to WWTW catchments and external catchment characteristics?
4. Which external variables show the clearest isolated associations with β , λ , duration and frequency?
5. When the external variables are considered together in multivariable supervised regression, how accurately can each outcome be predicted at an unseen WWTW, and which measured variables carry the most useful predictive information?

1.4 Contribution and scope

The project combines three methodological ideas: event-level EDM data rather than only annual summaries; sensor-level long-duration distribution parameters rather than a single national distribution; and a common geospatial predictor matrix linked to WWTW sewersheds. The study is observational. Statistical or predictive association is evidence of structure, not proof of a causal hydraulic pathway.

From spill events to catchment-scale prediction

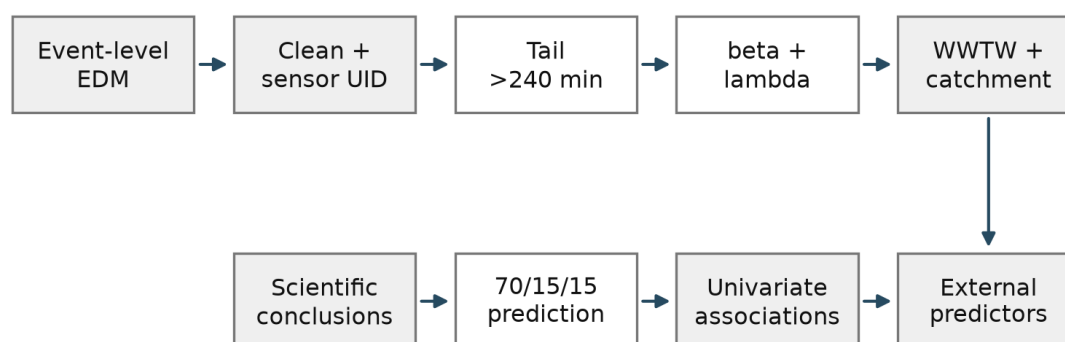


Figure 2: End-to-end analytical workflow from event-level EDM records to catchment-scale prediction.

The research question can only be answered if the event data from different water companies are first placed on a common, auditable footing. The next section therefore starts with acquisition and harmonisation rather than modelling.

2 From water-company records to an authoritative sensor master

2.1 Data acquisition

Event-level spill records were assembled across the participating English water companies using supplied EDM files and Environmental Information Regulation requests where required. The common analytical record contains a location identifier, permit or asset identifier, event start time, event end time and duration in minutes.

[TO COMPLETE: Insert a concise company-by-company data source and year-coverage table. Do not narrate every email or download. Summarise only differences that affect comparability.]

2.2 Cleaning and standardisation

Company-specific files were standardised into one event schema. Duration was calculated or checked in minutes from start and stop timestamps, with impossible or ambiguous rows retained in an audit path rather than silently coerced. This step was important because company formats and conventions differ, and errors in units or timestamp construction would propagate directly into the fitted duration distribution.

2.3 Sensor identity

Events were grouped into an authoritative sensor identity, preserving company, location name and permit or asset identifier. The national sensor master retained 25,409 unique sensors, of which 11,561 had accepted coordinates and 8,746 had a finite fitted β in the earlier national fitting release. A sensor remained in the master even if coordinates or a fitted tail parameter were unavailable, so missingness remained visible rather than becoming an invisible sample restriction.

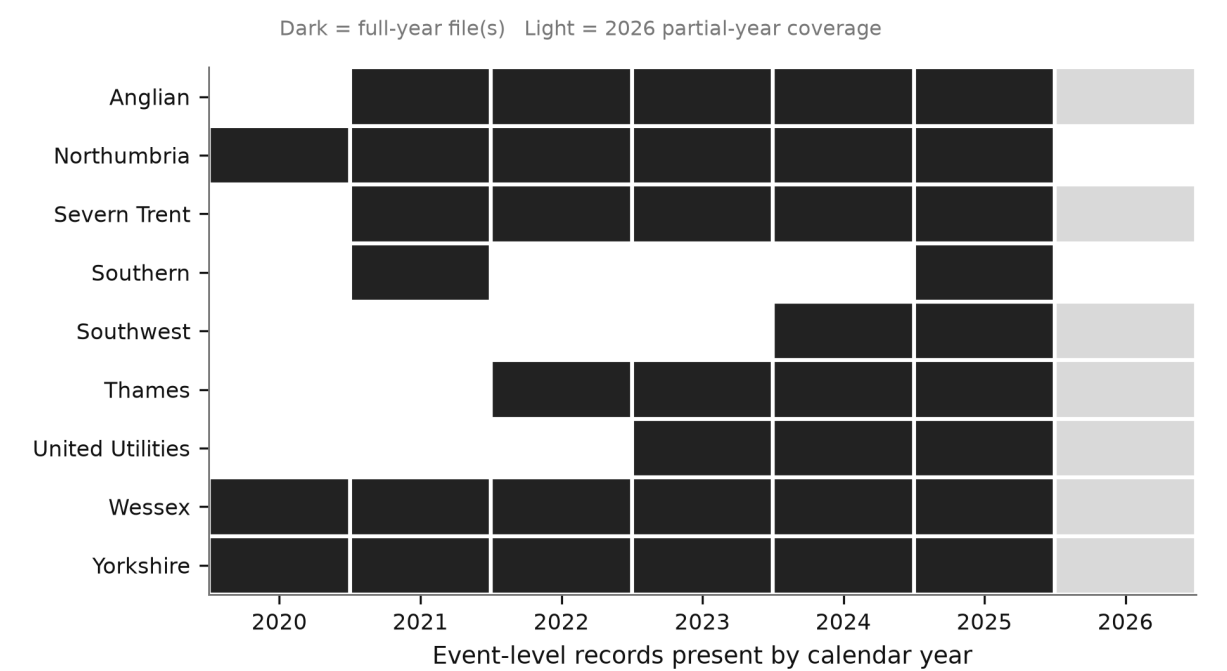


Figure 3: National EDM data coverage used to construct the event-level analysis.

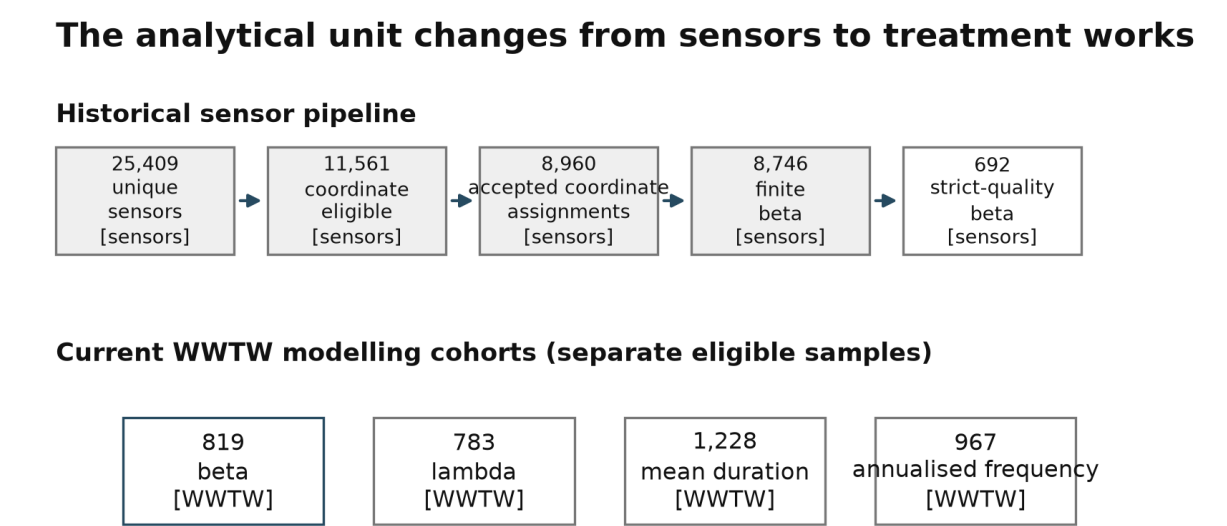


Figure 4: Project sample funnel from raw sensor identities to the final WWTW-level modelling cohorts.

Once the event records are comparable, the project can move from administrative records to a physical statistical object: the distribution of spill durations at each sensor.

3 The long-duration spill distribution

3.1 Why focus on events longer than four hours?

The active tail definition is fixed as

$$T > 240 \text{ minutes.} \quad (1)$$

The threshold separates the long-duration regime used for tail fitting from the complete event distribution. It is a design choice and must remain constant throughout the report.

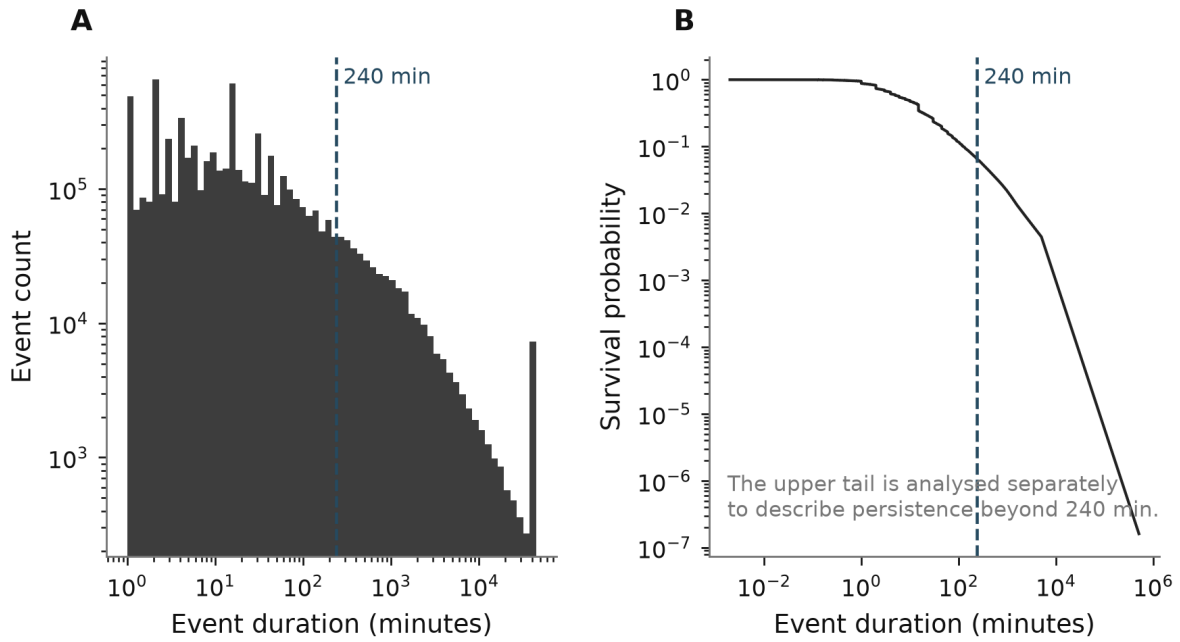


Figure 5: National spill-duration distribution and the long-duration analysis threshold.

3.2 Stretched-exponential model

For long-duration events, the fitted survival function is

$$S(t) = P(T > t) = \exp [-(\lambda t)^\beta], \quad (2)$$

where β controls tail shape and λ is an inverse timescale. At fixed λ , lower β produces a heavier, more persistent upper tail. At fixed β , higher λ shortens the characteristic timescale $1/\lambda$.

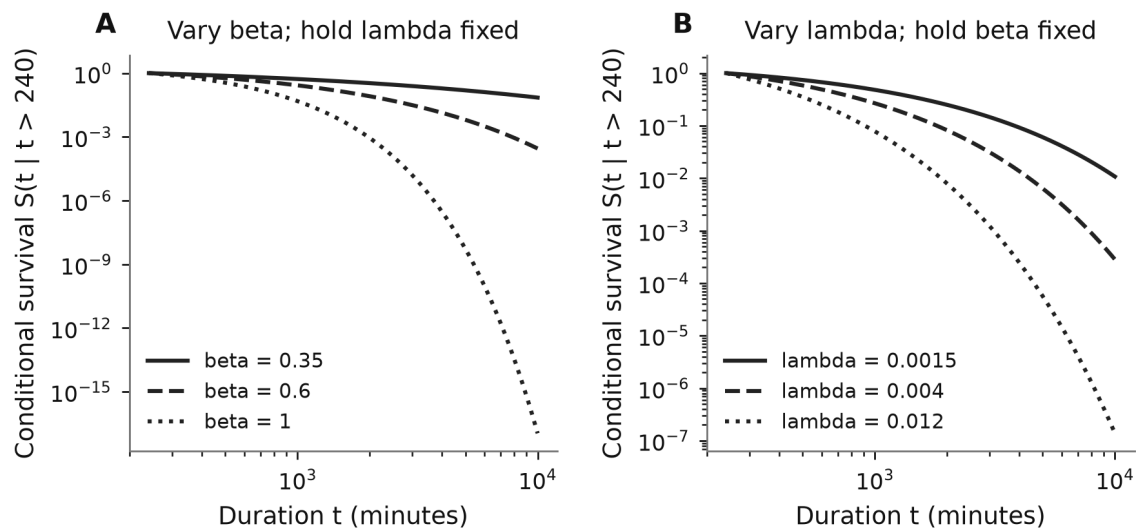
$$\beta < 1 \Rightarrow \text{stretched-exponential, persistent tail}, \quad \beta = 1 \Rightarrow \text{exponential}, \quad \beta > 1 \Rightarrow \text{compressed, faster-decaying tail.} \quad (3)$$

3.3 Sensor-level fitting and uncertainty

For each sensor with sufficient long-duration observations, the tail was fitted independently. Bootstrap resampling of observed tail events was used to describe parameter stability and to distinguish higher-quality estimates from broader finite fits. Bootstrap precision is not proof that the stretched-exponential model is a perfect physical representation; it describes uncertainty under the fitted empirical sample.

3.4 National parameter distributions

The broad WWTW-linked beta cohort contains 819 finite positive values in the current analysis, while the strict-quality sensitivity contains 227. The broad median beta is approximately 0.554 and the strict median approximately 0.355, placing the central tendency of both samples below the exponential reference $\beta = 1$.



Schematic under $S(t)=\exp[-(\lambda t)^\beta]$. Higher λ means a shorter characteristic time at fixed β .

Figure 6: How β and λ change the fitted survival curve.

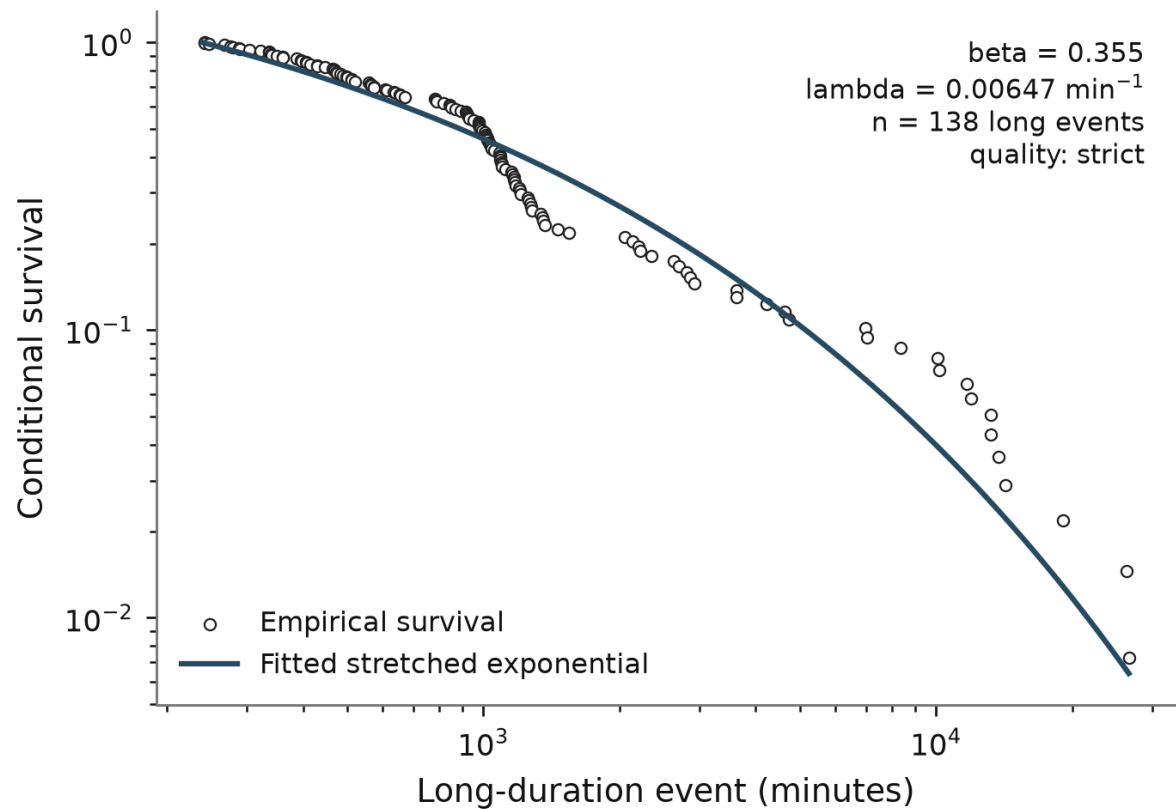


Figure 7: Example empirical long-duration survival curve and fitted stretched-exponential model for one representative sensor.

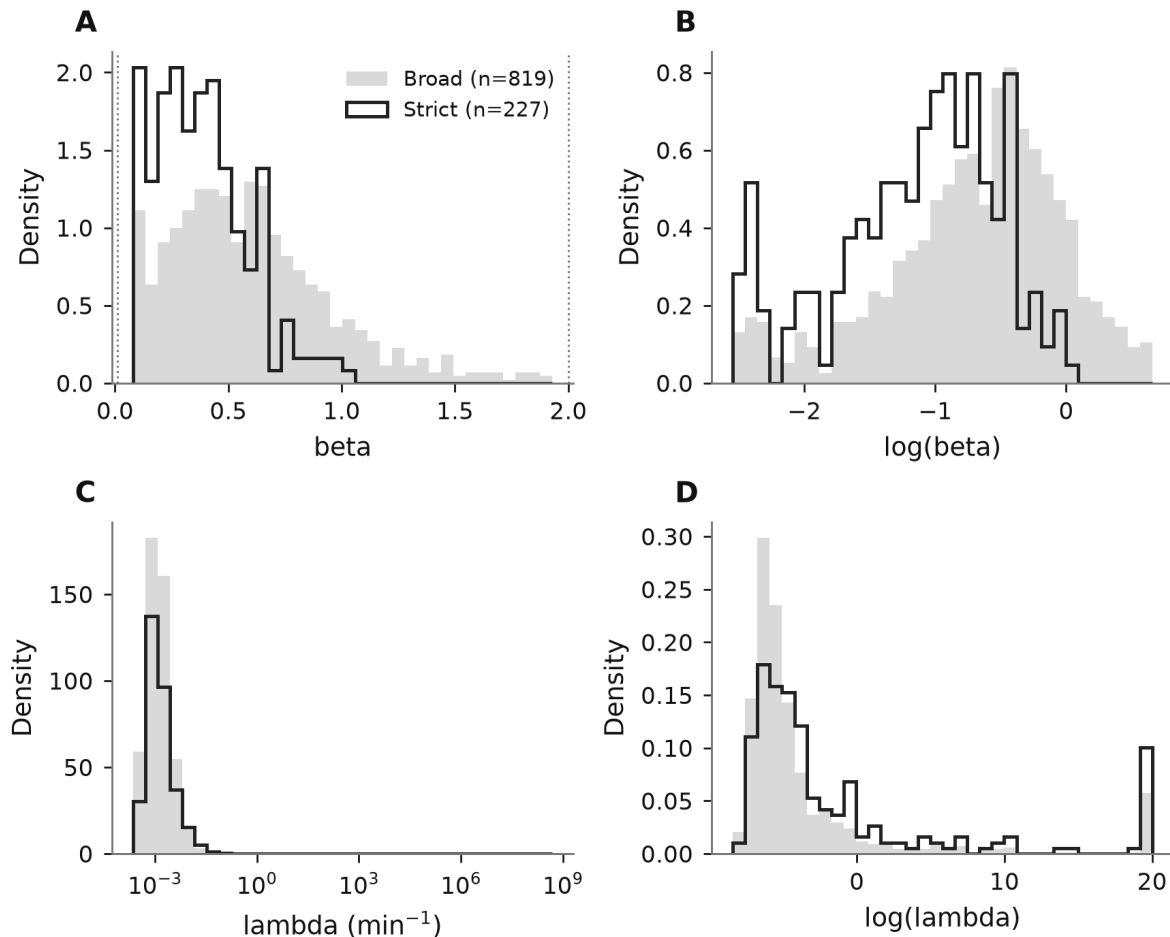


Figure 8: Distributions of fitted β and λ on modelling scales.

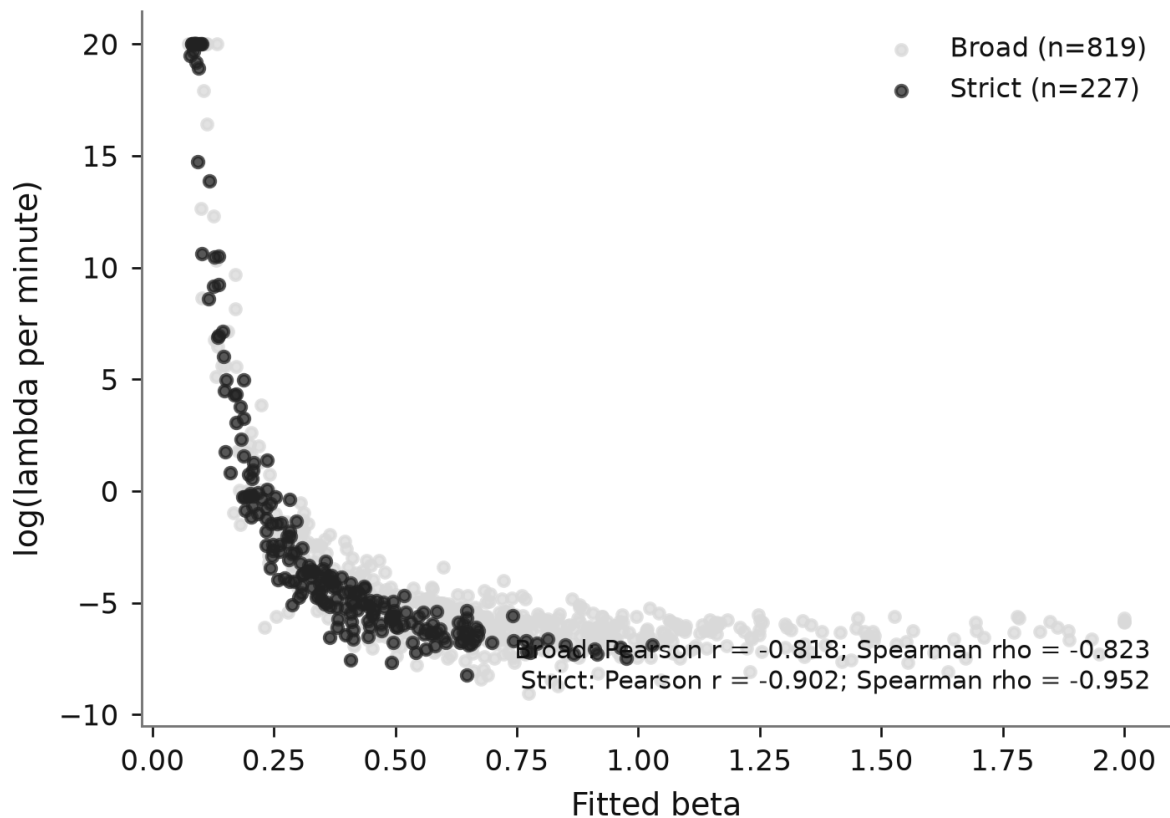


Figure 9: Joint relationship between fitted tail shape and inverse timescale.

The fitted parameters now exist at individual sensors, but the candidate explanations - rainfall, geology, population, capacity and land cover - exist at catchment or WWTW scale. The next challenge is therefore spatial identity: which sensor belongs to which wastewater system?

4 Linking CSO sensors to WWTW catchments

4.1 Wastewater catchments and WWTW sewersheds

Published wastewater catchment polygons were linked to treatment-work identifiers and dissolved by company and WWTW code to form system-scale sewersheds. These polygons provide the spatial unit over which capacity and environmental characteristics are attached.

4.2 Coordinate-based containment baseline

The first national assignment compared coordinate-eligible sensors in British National Grid (EPSG:27700) with same-company wastewater catchment polygons. Of 11,561 coordinate-eligible sensors, 8,960 were accepted through unique same-company containment or overlapping polygons that resolved to the same WWTW, an acceptance rate of approximately 77.5%.

4.3 Why a second WWTW-associated-sensor strategy was needed

The supervisor’s revised question focuses on overflow sensors plausibly associated directly with treatment works. A first fuzzy full-string name matcher was deliberately conservative but had low recall. Diagnostic review showed that facility suffixes such as “SSO”, “CSO”, “STW”, “WRC”, “storm tank” and “outfall” could dominate a similarity score even when the geographic place core was identical.

4.4 Place-core and supporting-evidence strategy

The refined strategy therefore separates geographic identity from facility terminology. For example, “SCAYNES HILL SSO” and “SCAYNES HILL STW” reduce to the shared place core “SCAYNES HILL”. Automatic matching remains same-company and beta-blind, preserves directional qualifiers such as NORTH and SOUTH, uses score-margin and ambiguity rules, and may use existing WWTW assignment, distance or regulatory outlet information as supporting evidence. Proximity alone is not treated as hydraulic proof.

Table 3: Evidence hierarchy for the final sensor-WWTW linkage.

Tier	Evidence
Regulatory-confirmed	Genuine regulatory storm-overflow or storm-tank identity linked to the WWTW, same company and no unresolved conflict.
Convergent high confidence	Strong place-core evidence plus one or more independent supports such as existing WWTW agreement, close outlet/site distance, sewershed consistency or a strong candidate margin.
Manually confirmed	Plausible but incomplete cases reviewed explicitly without inspecting beta or environmental outcomes.
Excluded / unresolved	Proximity-only candidates, cross-company conflicts, final-effluent-only sites or unresolved multiple-WWTW cases.

EVIDENCE NOTE Published catchment containment, names and public coordinates do not reconstruct the underground sewer network. Pumping, diversions and historical connections may be absent. The spatial assignment is a defensible analytical linkage, not a claim of exact pipe connectivity.

A defensible spatial link converts each fitted sensor outcome into a catchment-scale research unit. Only at this point is it meaningful to ask whether geology, rainfall, urban form or capacity correspond to systematic differences in spill behaviour.

5 Building the external catchment predictor matrix

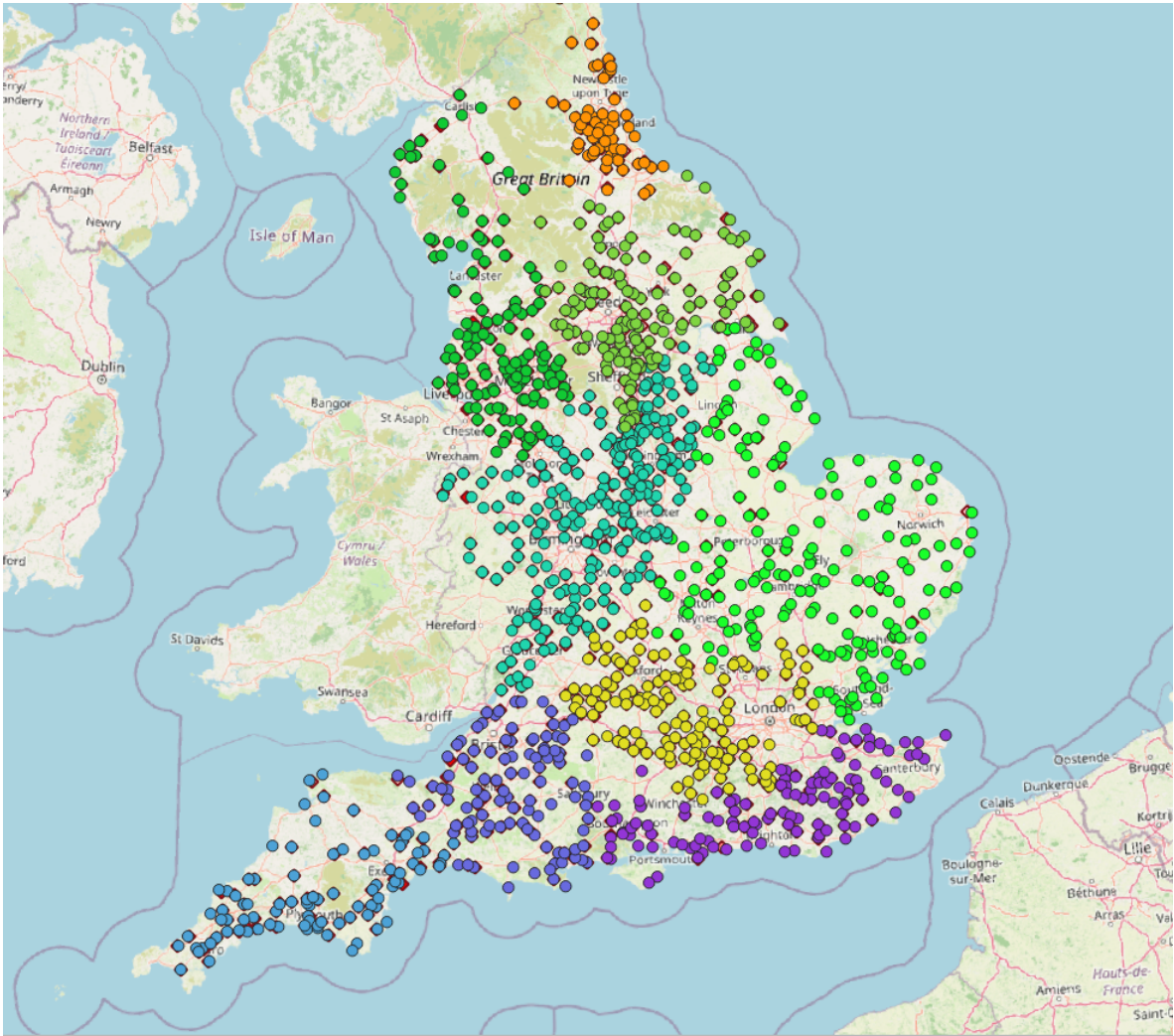


Figure 10: National spatial overview of WWTW catchments and associated CSO sensors.

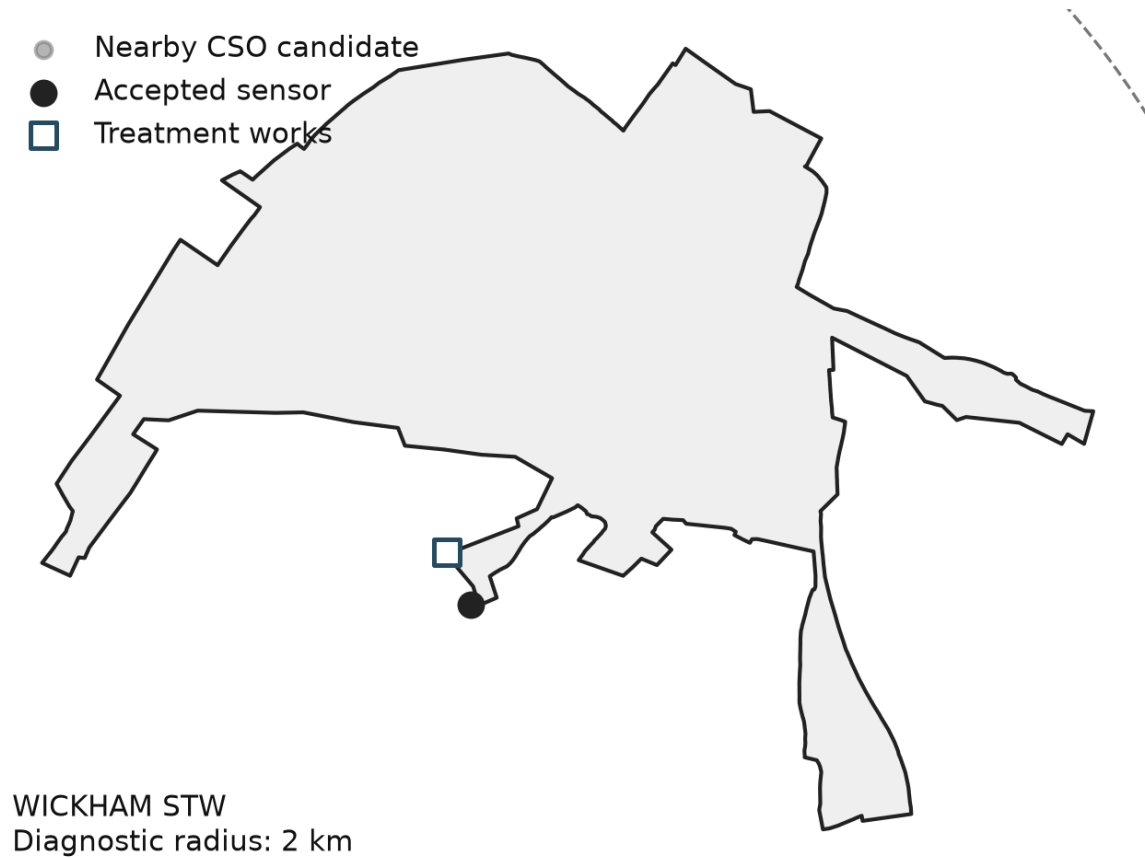


Figure 11: Example of the final sensor-to-WWTW linkage at local scale.

5.1 Treatment capacity

Treatment capacity utilisation is defined as

$$R_{\text{capacity}} = \frac{\text{load entering the WWTW}}{\text{reported design capacity}}. \quad (4)$$

The historic distribution has a median near 0.935, with many values exactly equal to 1 and none above 1. This may reflect reporting or censoring and is treated cautiously rather than as an exact wet-weather hydraulic headroom measure.

5.2 Hydrogeology

BGS Hydrogeology 625K polygons were intersected with each WWTW sewershed. For hydrogeological category k and sewershed j ,

$$H_{jk}(\%) = 100 \frac{A(C_j \cap G_k)}{A(C_j)}. \quad (5)$$

The percentage representation captures catchment composition rather than assigning a single point geology. Productivity shares and flow-mechanism shares are compositional, so exact redundant categories are not entered simultaneously in ordinary regression.

5.3 Rainfall

Two complementary rainfall descriptions are retained. Long-term climatic context is represented by static mean annual and winter rainfall. Monitoring-period variability is represented by annual NIMROD indices such as total wet-day precipitation, maximum 1-day and 5-day totals, rainfall intensity, heavy-rain days, consecutive wet days and consecutive dry days. Annual indices are area-weighted over the WWTW sewershed and require comparable monitoring-period coverage.

5.4 Urban form, population, age, terrain and scale

Land cover, population density, pre-1973 building share, mean terrain slope and WWTW sewershed area represent complementary aspects of runoff connectivity, wastewater loading, infrastructure age proxy, topographic response and system scale.

Table 4: Core external variables used in explanatory and predictive modelling. Extend this table only when a variable is scientifically distinct.

Variable family	Example field / representation	Unit	Why it may matter
Hydro productivity	high, moderate, low, no-groundwater shares	% area	Groundwater storage and transmission may alter infiltration/inflow persistence.
Hydro flow mechanism	intergranular, fracture-flow shares	% area	Distributed porous storage and preferential fracture pathways imply different hydraulic response timescales.
Static rainfall	mean annual; mean winter rainfall	mm	Long-term wetness, recharge and seasonal saturation context.
Annual NIMROD rainfall	PRCPTOT, Rx1day, Rx5day, SDII, R10, R20, CWD, CDD	mm / days	Monitoring-period rainfall amount, intensity and persistence.
Land cover	urban, suburban, continuous developed-land index	% / index	Imperviousness, connected drainage and urban hydraulic complexity.
Population	population density	people km ⁻²	Proxy for wastewater loading and urban system scale; not a direct causal mechanism by itself.
Infrastructure age proxy	pre-1973 building share	%	Approximate spatial proxy for older sewer infrastructure where direct sewer age is unavailable.
Terrain	mean slope	degrees	Affects runoff concentration, routing and pumping context.
Treatment operation	capacity utilisation ratio	ratio	Approximate loading relative to reported design capacity.
System scale	WWTW sewershed area	km ²	Routing distance, spatial averaging, storage and asynchronous subcatchment response.
Company context	water company category	category	Captures operator/region context; used as adjustment/sensitivity, not treated as a physical environmental variable.

The common predictor matrix now allows an initial question: what does each external variable appear to do on its own? That is useful for identifying patterns, but it cannot separate correlated mechanisms.

6 One-variable associations across four outcomes

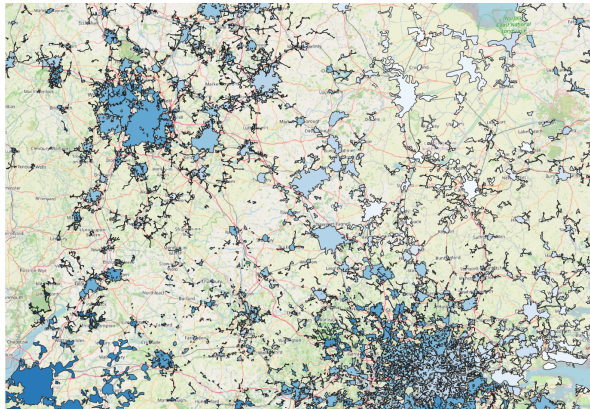
6.1 Method

For a positive outcome Y_i , one-variable models use

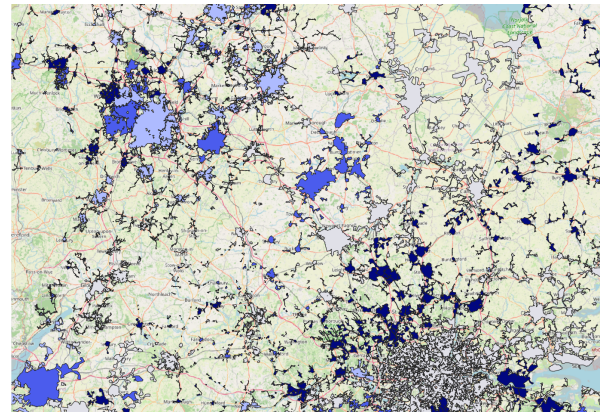
$$\log(Y_i) = \alpha + bX_i + \epsilon_i, \quad (6)$$

with HC3 robust covariance. Effects are reported as exact proportional differences for a meaningful predictor increment ΔX ,

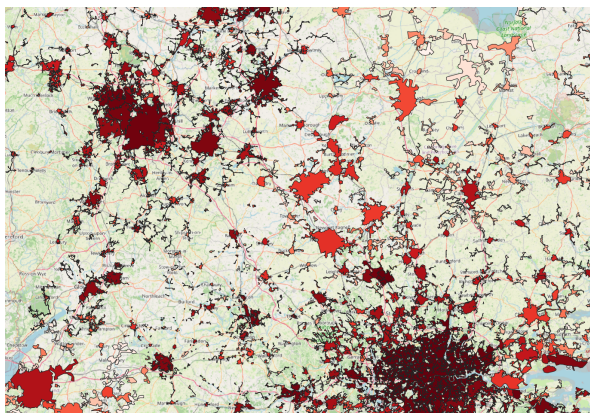
$$100 [\exp(b\Delta X) - 1] \%. \quad (7)$$



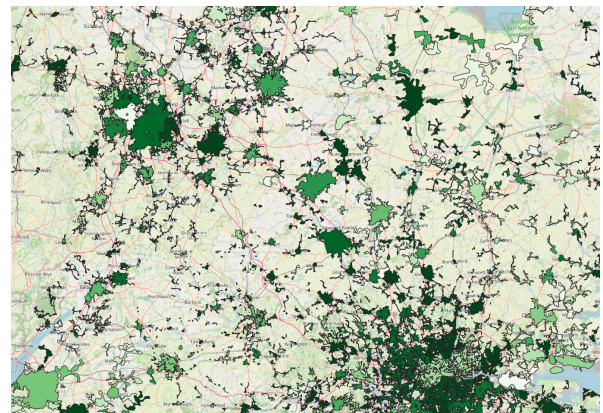
(a) Winter rainfall. Darker shading indicates greater mean winter rainfall.



(b) Hydrogeological productivity. Darker shading indicates higher productivity.



(c) Population density. Darker shading indicates more densely populated catchments.



(d) Capacity ratio. Darker green shading indicates a higher capacity ratio.

Figure 12: Spatial variation in four external predictor variables used in the analysis: (a) winter rainfall, (b) hydrogeological productivity, (c) population density, and (d) capacity ratio. Colour intensity represents the magnitude of each predictor within the mapped catchments.

All-variable Pearson correlation matrix

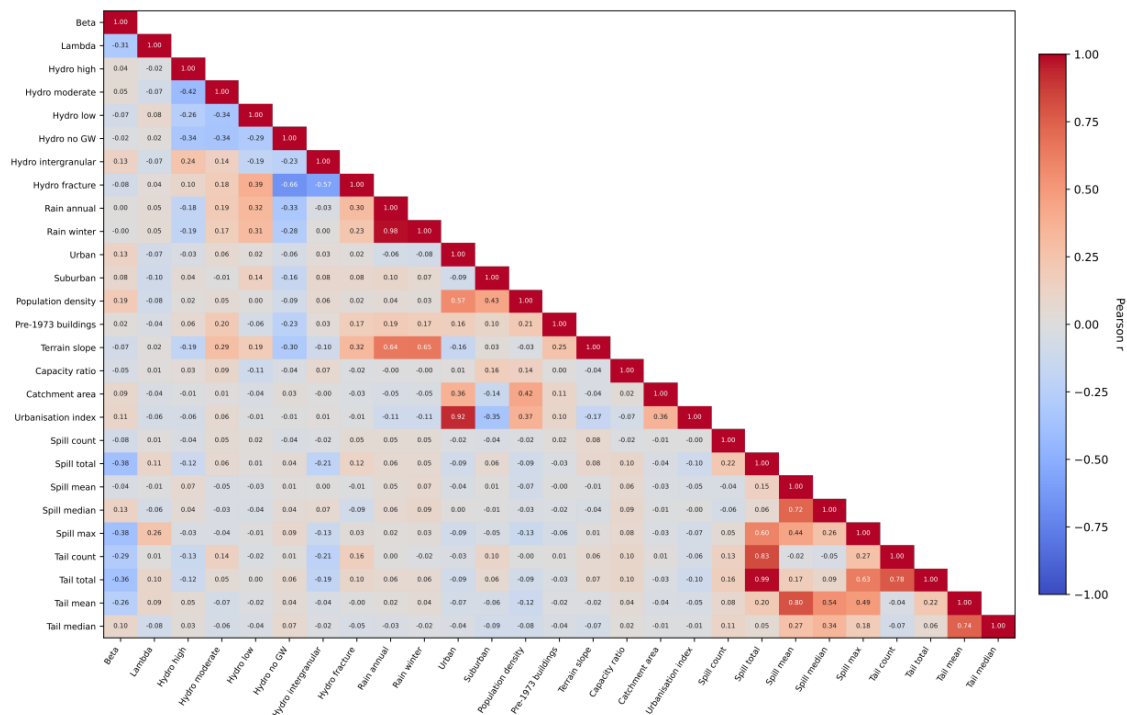


Figure 13: Correlation structure among the final external predictors.

Benjamini-Hochberg false-discovery-rate adjustment is applied within the pre-specified predictor families. These are unadjusted associations, not causal effects.

6.2 Tail shape β

The strongest unadjusted beta relationships are associated with urban-system structure and selected hydrogeological categories. In the strict-quality sample, a doubling of population density corresponds to approximately 23.6% higher beta, while a 10-percentage-point increase in urban land cover corresponds to approximately 23.1% higher beta. Low-productivity and fracture-flow hydrogeology correspond to lower beta, whereas intergranular-flow hydrogeology corresponds to higher beta. Static annual and winter rainfall show negative beta associations in the strict sample, consistent with a heavier long-duration tail in wetter settings, but those rainfall signals weaken in the broader sample.

6.3 Inverse timescale λ

The lambda analysis reveals a useful complement to beta. Low-productivity and fracture-flow settings tend toward higher lambda, while moderate-productivity and intergranular settings tend toward lower lambda in the strict analysis. Urbanisation and population density are associated with substantially lower lambda. Because beta and lambda are jointly estimated and lambda can be extremely variable, these results should be interpreted as parameter regimes rather than as isolated deterministic effects.

6.4 Mean spill duration and annualised frequency

Mean spill duration is more strongly structured by the measured external covariates than annualised frequency in the univariate stage, although even the largest single-predictor R^2 values are small. Suburban land cover and treatment-capacity utilisation show notable positive duration associations. Static winter rainfall and sustained multi-day annual rainfall also correspond to longer mean duration, while longer consecutive dry spells correspond to shorter duration. Frequency shows clearer hydrogeological and catchment-scale structure; intergranular-flow share and larger catchment area are associated with lower annualised frequency, and no annual NIMROD rainfall index survives the rainfall-family FDR screen.

6.5 Cross-outcome synthesis

The one-variable stage is scientifically useful because it shows that the four outcomes are not interchangeable. Variables that correspond to a heavier tail do not necessarily produce longer mean events or greater frequency. It also

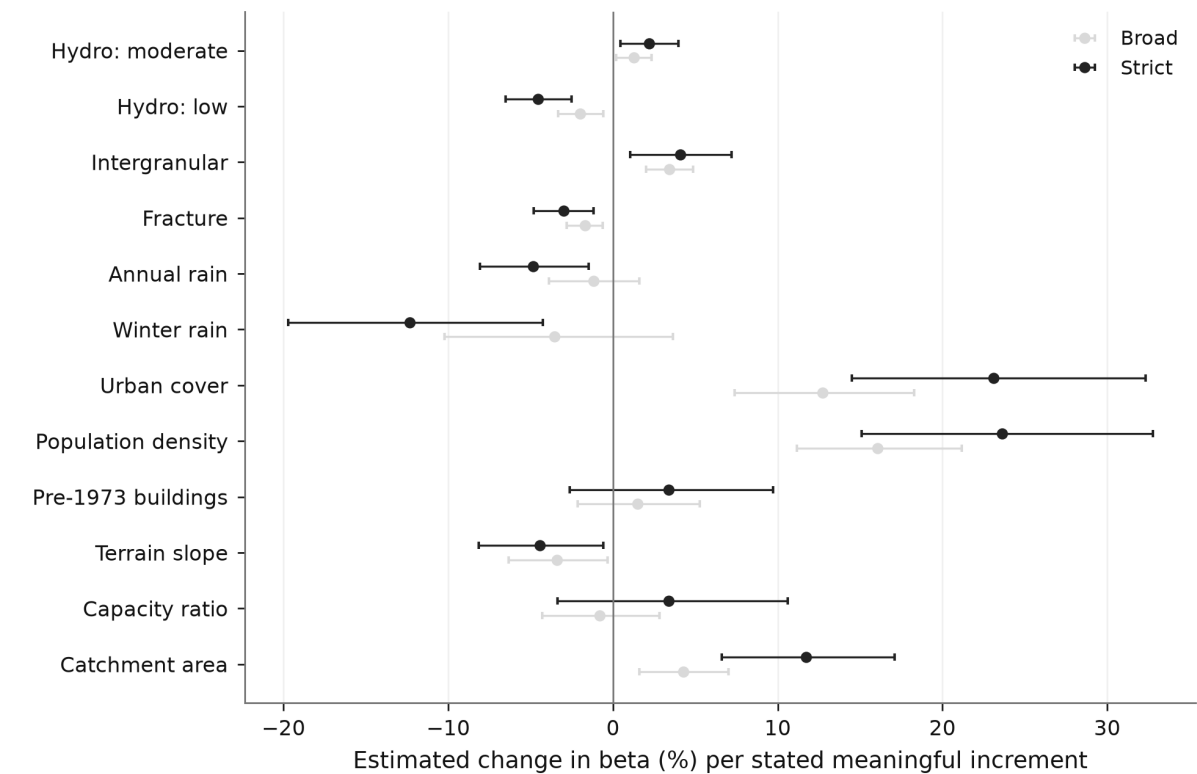


Figure 14: Unadjusted effects on fitted β .

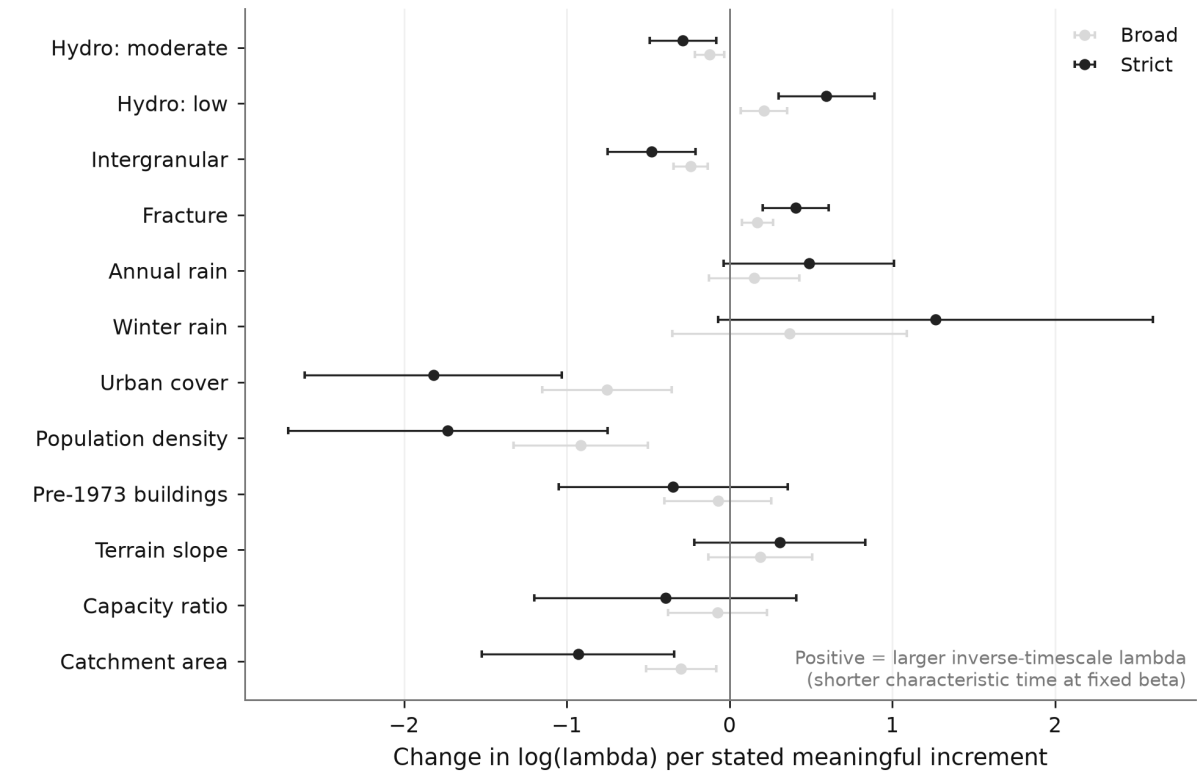


Figure 15: Unadjusted effects on fitted inverse-timescale λ .

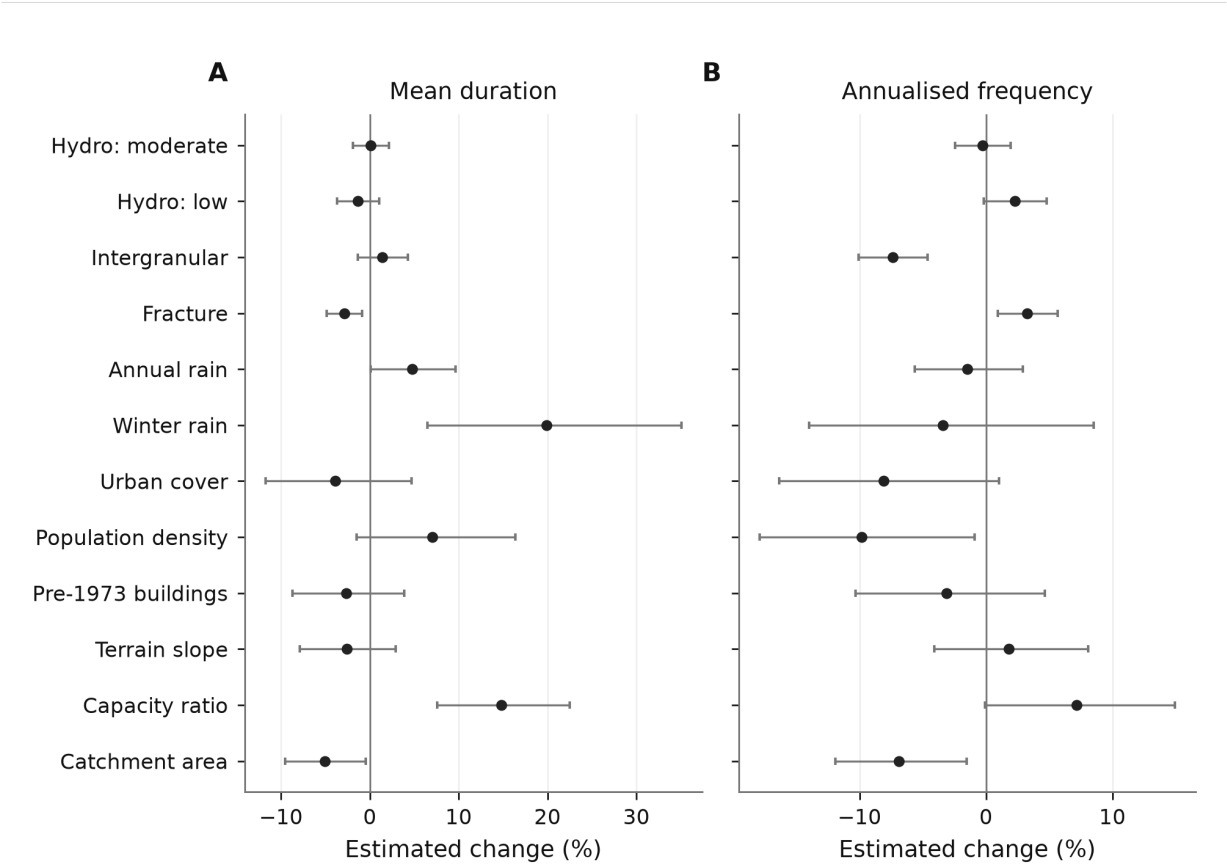


Figure 16: Unadjusted effects on mean spill duration and annualised spill frequency.

reveals the main problem that motivates adjustment: rainfall variables overlap with each other, urbanisation overlaps population and catchment scale, and hydrogeology percentages are compositional.

Table 5: Concise one-variable evidence synthesis. Keep the main text to the strongest patterns; place the complete coefficient atlas in the appendix.

Mechanism	Main observed pattern	Interpretation to carry forward
Urban system	Higher urban cover / population generally $\beta \uparrow$, $\lambda \downarrow$	Longer characteristic timescale but smoother/faster-decaying relative tail; strong overlap among urban proxies requires adjustment.
Hydrogeology	Intergranular/moderate and fracture/low-productivity often occupy opposite β - λ regimes	Flow architecture appears informative, but percentages are compositional and spatially structured.
Rainfall	Wetter winter climate tends toward $\beta \downarrow$; duration responds to winter and multi-day rainfall	Hydrological state and persistence may matter more than isolated intensity for long-duration behaviour.
System scale	Catchment area and capacity show outcome-specific associations	Operational and geometric scale can confound environmental patterns and must be adjusted jointly.

The one-variable stage shows where potentially meaningful relationships sit, but each predictor is still being examined in isolation. The final analytical question is deliberately harder: when the full external catchment profile is considered together, does it contain enough information to predict an unseen WWTW? The report therefore moves directly from univariate evidence to the supervisor-directed multivariable machine-learning programme.

7 Predicting unseen WWTWs: the final multivariable machine-learning analysis

7.1 Why this is the main final analysis

The univariate stage asks whether one catchment characteristic is associated with one outcome when examined by itself. The final machine-learning stage asks a more demanding and operationally relevant question: *if all available external information is supplied simultaneously, can a regression model predict the outcome at a WWTW that it has not seen before?* This is the project’s main multivariable analysis. It uses scikit-learn regression models, a locked holdout design and RMSE as the headline measure of predictive performance.

The four outcomes remain separate because they describe different physical properties and have different amounts of usable data. Forcing them into one common multi-output model would discard valid rows and blur fundamentally different statistical targets.

Table 6: Four separate prediction problems used in the final analysis.

Model	Target	Physical question
Model 1	$\log(\beta)$	Can catchment conditions predict the shape of the long-duration spill tail?
Model 2	$\log(\lambda)$	Can catchment conditions predict the inverse characteristic timescale of long spills?
Model 3	Mean spill duration	Can catchment conditions predict how long a typical valid spill lasts?
Model 4	Annualised spill frequency	Can catchment conditions predict how often valid spills occur per monitored year?

7.2 The 70/15/15 protocol

For each target, approximately 70% of WWTWs are used for training, 15% for validation and 15% for a locked test. Preprocessing, imputation and standardisation are fitted inside scikit-learn pipelines. Ridge and Elastic Net are tuned within the training portion; the validation set selects the frozen candidate; train plus validation are then refitted before the test set is evaluated once.



Figure 17: Supervisor-directed prediction protocol. The locked test is not used to choose algorithms, predictors or hyperparameters.

7.3 What RMSE tells us

The headline metric is root mean squared error,

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (8)$$

Lower RMSE means smaller prediction error, with large errors receiving extra weight because they are squared. Every fitted model is compared with a Dummy mean predictor. The percentage improvement is

$$100 \frac{\text{RMSE}_{\text{Dummy}} - \text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{Dummy}}}. \quad (9)$$

A 7% improvement does *not* mean 93% accuracy. It means that the model reduces RMSE by 7% relative to predicting one constant value for every unseen WWTW. Bootstrap uncertainty, R^2 , Spearman rank correlation and robustness checks are therefore interpreted alongside RMSE.

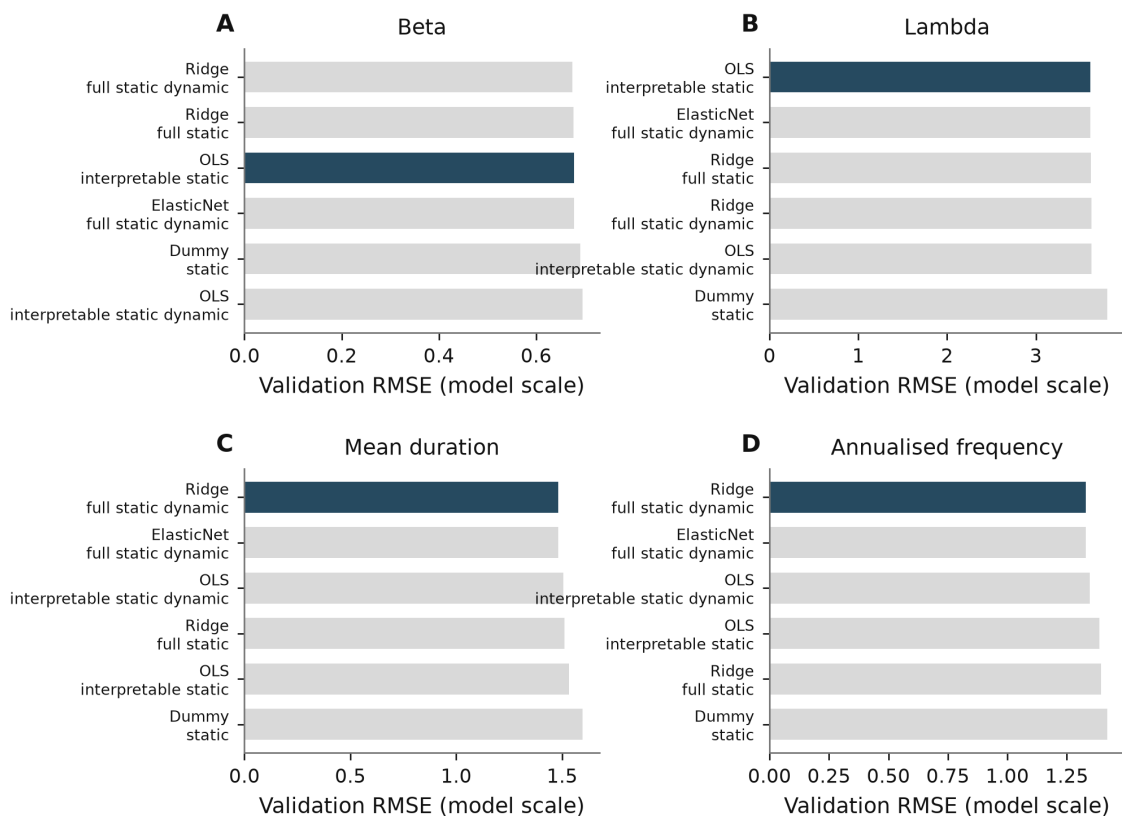
7.4 Overall predictive scorecard

7.5 How to read the predicted-versus-observed and residual figures

The predicted-versus-observed plots are the most direct visual test of whether the model reproduces individual sites. The horizontal axis is the observed value and the vertical axis is the model prediction. A perfect model would place every point on the 1:1 diagonal. A cloud compressed toward the centre means the model has learned some broad ordering but is shrinking unusual WWTWs back toward typical values. This centre-compression is visible for all four outcomes and is especially severe for extreme lambda and very long-duration sites.

Table 7: Final 70/15/15 predictive results. RMSE values are target-specific and should not be compared directly between outcomes.

Target	Eligible n	Test n	Selected model	RMSE gain	R^2	Interpretation
Beta	819	123	OLS, interpretable static	4.9%	0.073	Modest / qualified signal; bootstrap improvement includes zero.
Lambda	783	118	OLS, interpretable static	4.9%	0.095	Weak / inconclusive; extreme fitted values and target instability dominate.
Mean duration	1,228	185	Ridge, static + dynamic	2.6%	0.052	Modest / qualified; largest sample does not translate into strong prediction.
Annualised frequency	967	146	Ridge, static + dynamic	7.0%	0.133	Strongest of the four, but still modest site-level predictive skill.

**Figure 18:** Validation RMSE comparison used to select the final model for each target.

Residual plots show the remaining error after prediction. Residuals should ideally scatter around zero without a systematic curve or widening fan. Large tails identify WWTWs whose behaviour is not captured by the current external feature bank. Those difficult sites are retained because they are part of the scientific limitation rather than nuisance points to be removed.

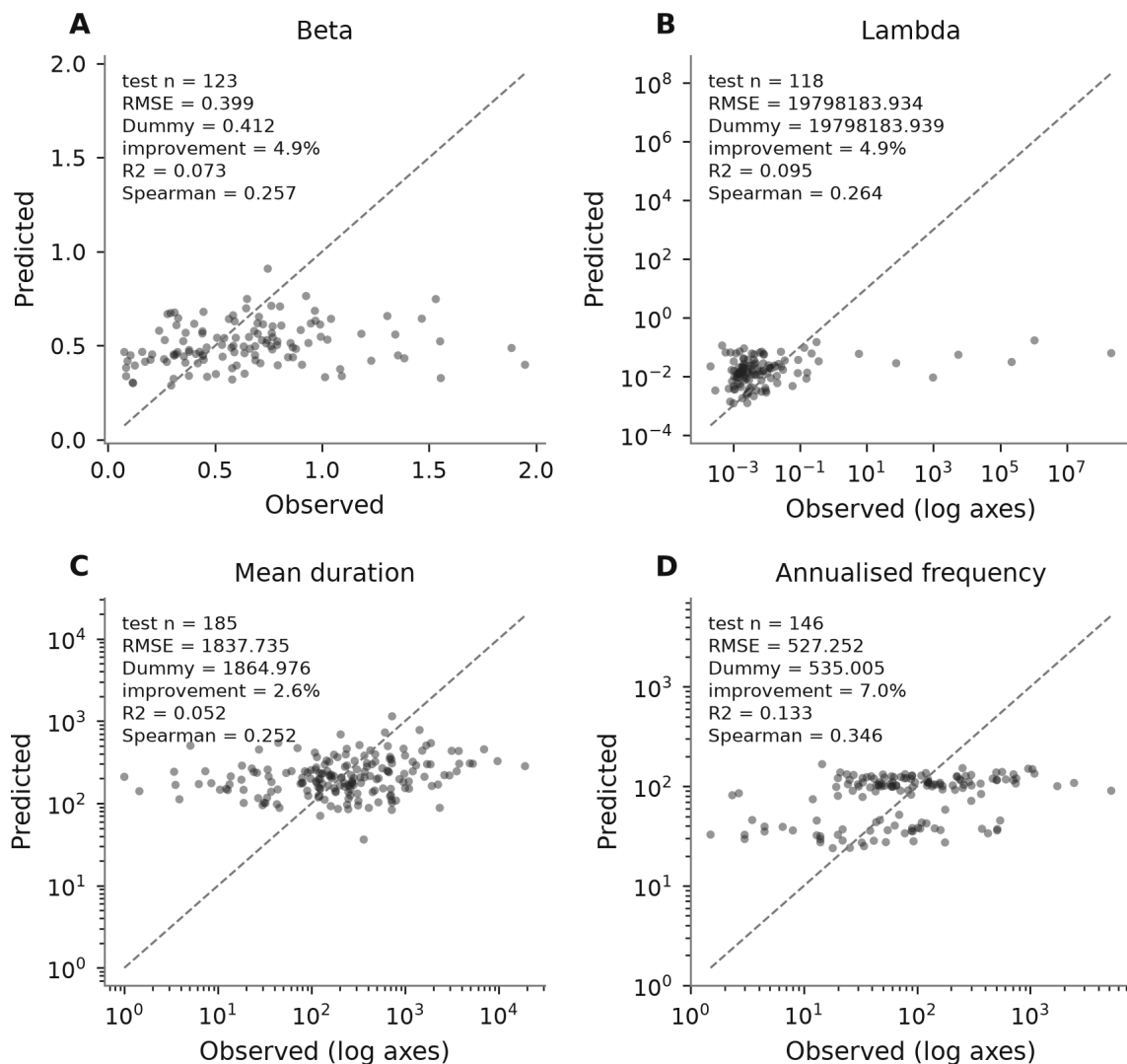


Figure 19: Locked-test predicted-versus-observed diagnostics for the four final models.

7.6 Model 1: predicting the fitted tail-shape parameter β

EVIDENCE NOTE Predictive signal: modest and qualified. Test RMSE is 0.653 compared with 0.687 for the Dummy model, a 4.9% reduction. Test $R^2 = 0.073$ and Spearman correlation is 0.257. The bootstrap improvement interval $[-1.0\%, 10.3\%]$ crosses zero, so the point improvement is not sufficiently stable to describe as strong predictive skill.

The model therefore contains some information about which unseen WWTWs tend to have higher or lower beta, but it reproduces only a small fraction of the observed site-to-site variation. Predictions remain concentrated near typical beta values and tend to miss high-beta sites. Population density, intergranular-flow geology and low-productivity geology are among the measured features most relied upon by the fitted predictor system. Annual dynamic rainfall provides only a small improvement.

What this could mean. Beta is a fitted property of the long-duration tail rather than a directly observed hydraulic state. Its predictability is therefore limited both by missing explanatory information and by uncertainty in the fitted beta label itself. A weak predictive result is compatible with real univariate environmental associations if much of beta's remaining variance depends on unmeasured network or event conditions.

Most useful missing information. Higher-quality tail fits and joint beta-lambda uncertainty; direct sewer storage and network topology; wet-weather hydraulic headroom; groundwater level and soil saturation; event-aligned antecedent

rainfall; pumping and control state; and information on changes in operation or maintenance during the monitoring period.

7.7 Model 2: predicting inverse-timescale λ

EVIDENCE NOTE **Predictive signal: weak / inconclusive.** Test RMSE is 3.855 compared with 4.056 for the Dummy model, again a 4.9% point improvement, with test $R^2 = 0.095$. The bootstrap improvement interval $[-2.3\%, 8.6\%]$ crosses zero.

Lambda is the least stable predictive target. A small number of fitted values are orders of magnitude larger than the main population, and the selected model cannot reproduce those extremes. The resulting residual distribution is highly sensitive to which extreme WWTWs appear in a fold or holdout. Dynamic annual rainfall does not rescue this target and slightly worsens the pre-specified comparison.

What this could mean. The limitation may sit partly in the target rather than purely in the predictors. Beta and lambda are estimated jointly from the same tail distribution, so parameter covariance and identifiability can make lambda noisy even when the underlying spill process is meaningful. This makes increasingly aggressive algorithms a poor first response.

Most useful missing information. Joint beta–lambda bootstrap covariance, improved parameter-identifiability diagnostics, event-mixture descriptors, direct hydraulic state, storm storage, network travel/storage characteristics, groundwater infiltration, pumping/control state and event-specific rainfall history.

7.8 Model 3: predicting mean spill duration

EVIDENCE NOTE **Predictive signal: modest and qualified.** Test RMSE is 1.631 compared with 1.675 for the Dummy model, a 2.6% reduction, with test $R^2 = 0.052$ and Spearman correlation 0.252. The bootstrap improvement interval $[-1.5\%, 6.5\%]$ crosses zero.

This model has the largest cohort, 1,228 WWTWs, but the smallest point improvement over the baseline. That matters because it shows that sample size alone is not the main obstacle. Static winter rainfall, static annual rainfall and annual SDII are among the most relied-on measured features, while adding company identity produces a much larger RMSE reduction than it does for the other targets.

What this could mean. Mean duration appears to depend strongly on operator/region-specific or hydraulic information that is not explicitly represented by the current catchment variables. Company should not be interpreted as causal; it is acting as a proxy for combinations of sewer design, operational practice, reporting, maintenance and regional physical conditions.

Most useful missing information. Storm-storage volume, FFT and wet-weather treatment limits, network topology and pipe/storage capacity, pumping state, maintenance/control history, groundwater infiltration, event-level antecedent wetness, storm intensity and duration, and more direct information on combined/separate sewer structure.

7.9 Model 4: predicting annualised spill frequency

EVIDENCE NOTE **Predictive signal: the strongest of the four, but still modest.** Test RMSE is 1.341 compared with 1.442 for the Dummy model, a 7.0% reduction. Test $R^2 = 0.133$, Spearman correlation is 0.346 and the bootstrap improvement interval $[0.8\%, 12.1\%]$ remains above zero.

This is the clearest evidence that the current feature system adds reproducible information beyond a constant prediction. Intergranular-flow geology, annual consecutive dry days and population density are among the measured features most relied upon. Annual dynamic rainfall also gives its largest improvement for this target. However, predictions still compress the highest-frequency sites toward the centre, so “strongest” does not mean “accurate”.

What this could mean. Frequency is a more direct observed outcome than beta or lambda and may therefore contain less parameter-fitting noise. Its stronger response to annual rainfall is physically plausible because rainfall controls event opportunity, but the modest R^2 shows that rainfall and static catchment form are still far from a complete description of overflow occurrence.

Most useful missing information. More precise monitored exposure, harmonised reporting rules, event-level rainfall opportunity, sewer surcharge state, storage availability, pump/control operation, dry-weather loading, hydraulic headroom and temporary operational changes.

7.10 Which measured variables does each model rely on?

The final models are multivariable, so variable impact should be interpreted conditionally within the full predictive system. A useful ranking is the increase in validation or cross-validated RMSE when a variable is removed and the model is refitted. This is complemented by standardised coefficient direction, coefficient sign stability and permutation importance. Correlated predictors can share or redistribute importance, so these rankings are predictive rather than causal.

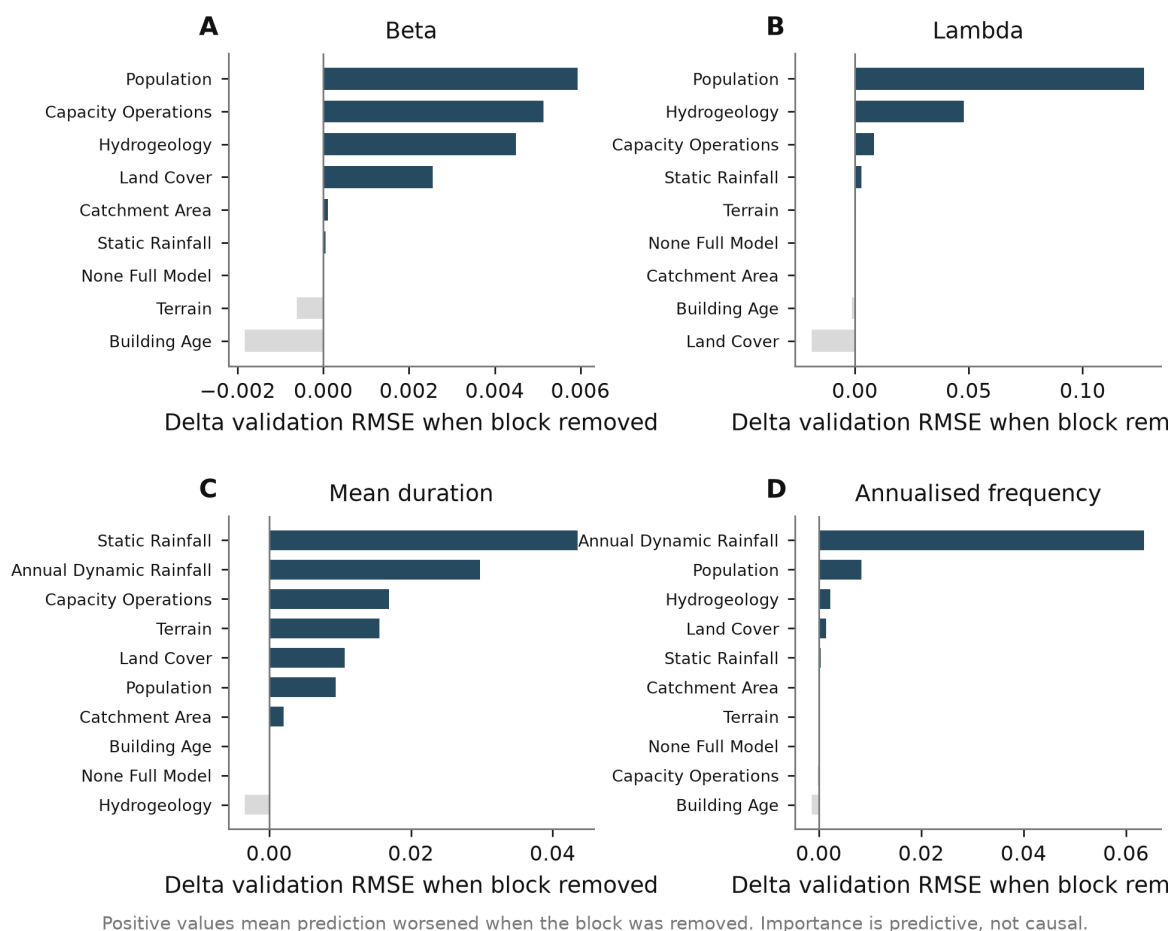


Figure 20: Predictive importance of the measured catchment variables across beta, lambda, duration and frequency.

7.11 Did annual NIMROD rainfall add useful information?

The static-versus-static-plus-annual-rainfall comparison asks whether monitoring-period annual rainfall adds information beyond the pre-existing static feature bank. It reduces test RMSE by approximately 0.015 for beta, 0.011 for duration and 0.054 for annualised frequency, but increases lambda-model RMSE by approximately 0.101. The result is therefore incremental rather than transformative: annual rainfall helps some outcomes, particularly frequency, but it does not solve the missing-information problem.

7.12 Learning curves: are the models limited mainly by the amount of data?

Figure 22 is a diagnostic of whether collecting more WWTWs with the *same type of information* is likely to produce a large performance gain. The x-axis is the number of WWTWs used for training and the y-axis is RMSE on the model scale. The solid black curve is error on the data used to fit the model. The dashed grey curve is cross-validation error on held-out folds.

Training error often rises as more WWTWs are added because a small model can fit a small, less diverse sample unusually closely. Cross-validation error should fall if additional observations improve generalisation. The most important features are therefore the *gap* between the curves and whether the cross-validation curve is still falling at the right-hand edge. Absolute RMSE values must not be compared between panels because each target is on a different scale.

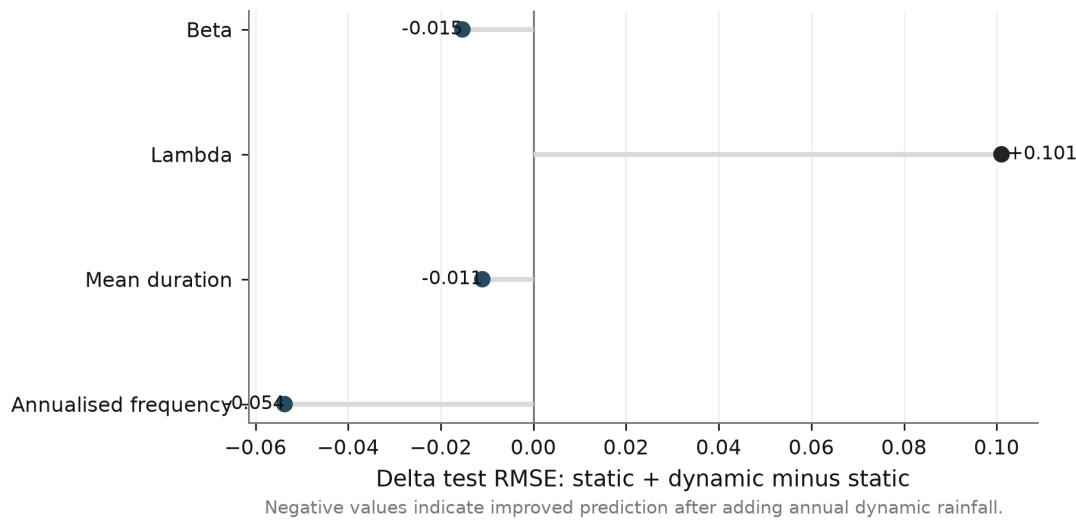


Figure 21: Change in prediction RMSE after adding the annual NIMROD rainfall block.

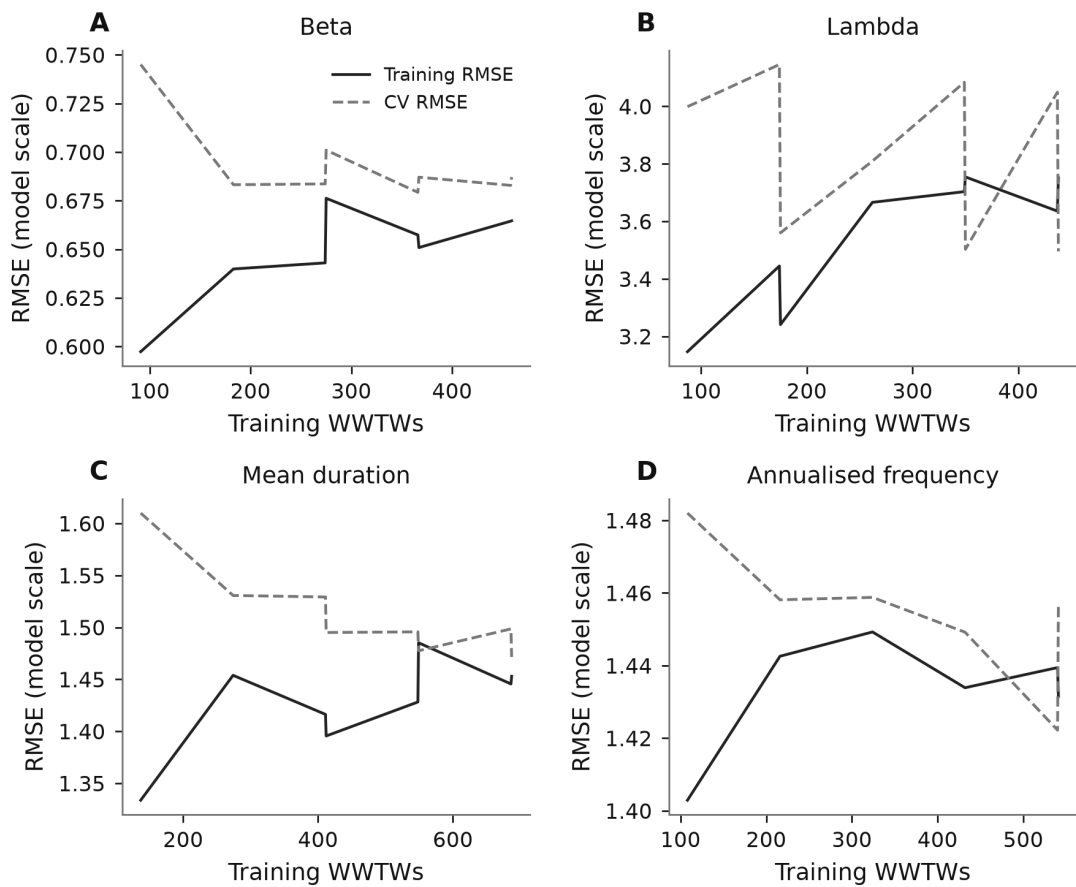


Figure 22: Learning curves for the four prediction targets. Solid black lines show training RMSE; dashed grey lines show cross-validation RMSE. The curves diagnose whether more WWTWs of the same information type are likely to reduce prediction error substantially.

Table 8: Interpretation of the four learning-curve panels.

Target	What the curves show	Research implication
Beta	Cross-validation RMSE drops markedly at first, then settles around a similar level while the training–CV gap becomes small.	The model is approaching an information-limited plateau. More comparable beta-labelled WWTWs may improve precision, but a large jump in predictive skill is unlikely without better beta labels or richer hydraulic/temporal variables.
Lambda	The cross-validation curve is irregular rather than smoothly improving, with large fold-to-fold jumps as training size changes.	Performance is sensitive to which WWTWs enter the folds, consistent with the extreme and unstable lambda distribution. Target reliability and joint beta–lambda identifiability should be improved before interpreting this as a simple sample-size problem.
Mean duration	With the largest sample, training and cross-validation errors move close together and the CV curve largely flattens.	This is the strongest evidence that “more of the same rows” is not the main solution. Missing hydraulic, operational and event-state information is the more plausible bottleneck.
Annualised frequency	Cross-validation RMSE declines more consistently than for the other targets and the gap narrows, but improvement also begins to flatten at the largest sample sizes.	Additional diverse, well-monitored WWTWs may still help somewhat, but the current 7% baseline gain is already approaching a modest-skill plateau. Better exposure and event-opportunity variables remain important.

EVIDENCE NOTE Overall learning-curve conclusion. The curves do not support the idea that a much larger sample of otherwise identical WWTW rows will suddenly create high predictive accuracy. Beta and duration are close to plateaus, lambda is unstable rather than cleanly data-limited, and frequency still benefits somewhat from additional data but is also flattening. The next major information gain should therefore prioritise better target measurement and variables closer to the physical sewer state.

7.13 What the final machine-learning stage tells us scientifically

The four models contain genuine but limited predictive information. Frequency is the clearest target, beta contains a qualified weak signal, duration remains weak despite the largest sample and lambda is dominated by target instability. The common pattern is centre-compression: models reproduce typical sites more easily than unusual extremes. This is exactly what would be expected when the available predictors describe broad catchment context but omit the instantaneous hydraulic and operational state that determines whether a particular system behaves exceptionally.

The final predictive analysis does not overturn the earlier environmental relationships. It places a quantitative ceiling on how much those measured relationships can currently do. The project therefore ends not with a claim of high-accuracy prediction, but with a clearer physical diagnosis of what information the present catchment-scale dataset captures and what remains missing.

8 Integrated discussion

8.1 From association to mechanism

The strongest result is not one universal environmental coefficient. It is the emergence of distinct response regimes. Urbanisation and population are associated with higher beta and lower lambda in the univariate parameter analysis, suggesting a longer characteristic timescale paired with a more regular relative tail. Intergranular and fracture-flow hydrogeology often occupy opposing beta-lambda regimes, which is more informative than treating aquifer productivity as a single monotonic scale. Rainfall appears more important for mean duration and for hydrological state than as a universal direct predictor of fitted tail parameters.

8.2 Why beta, lambda, duration and frequency must remain separate

The four outcomes encode different parts of spill behaviour. A catchment can have a heavy long-duration tail without high annual event frequency. A variable can alter the timescale of an event distribution without changing its relative tail shape in the same direction. The paired beta-lambda interpretation therefore prevents simplistic statements such as “higher rainfall means longer spills” when the evidence is actually about tail persistence.

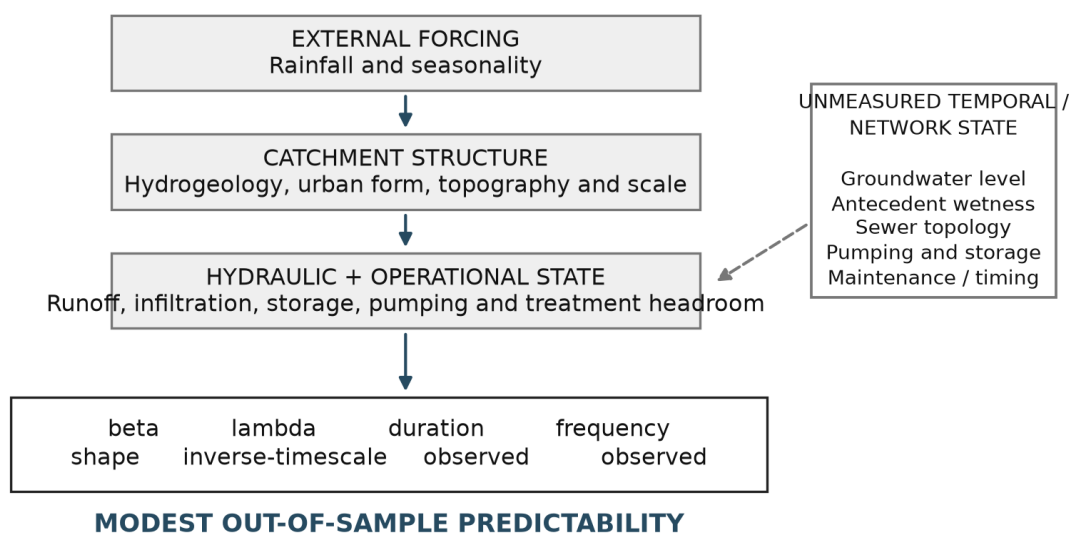
8.3 Explanation is not prediction

The one-variable regressions identify isolated environmental relationships, whereas the final machine-learning models ask whether the full predictor profile can forecast an unseen WWTW. The modest test-set RMSE improvements show why the distinction matters: a clear isolated association can coexist with weak whole-system prediction when most site-to-site variation depends on information that is not present in the feature bank.

8.4 What the machine-learning result adds to the scientific story

A weak or modest model is not a failed final chapter. It quantifies the information ceiling of the measured feature set. The external environment matters, but the remaining prediction error points toward omitted hydraulic state: sewer topology, storage, pumping, maintenance, groundwater level, antecedent wetness, storm timing, treatment control and other time-varying operational conditions.

What the project learned



Solid arrows summarise the analytical pathway; dashed arrows denote hypothesised mechanisms.

Figure 23: Integrated scientific synthesis of the project from external catchment context to observed CSO outcomes and the remaining unmeasured hydraulic state.

9 Limitations

1. **Hydraulic connectivity is inferred, not observed.** Published sewersheds, names and outlet coordinates do not fully reconstruct underground network topology.
2. **WWTW-associated sensors are a selected population.** Results may not generalise to every upstream network CSO.
3. **Beta and lambda are estimated responses.** Their reliability varies with the number and structure of tail events, and lambda is especially unstable in some fits.
4. **Capacity is an imperfect operational proxy.** The observed cap at 1.0 suggests source semantics or censoring and does not directly measure storm-weather hydraulic headroom.
5. **Environmental variables are correlated and spatially structured.** Hydrogeology and land cover are compositional; rainfall metrics overlap strongly; company is geographically clustered.
6. **Annual dynamic rainfall remains an aggregate.** It is temporally closer to the monitoring period than climatology, but it still does not encode the rainfall preceding each individual spill.
7. **Predictive generalisation is modest.** Random/grouped holdouts are informative but do not fully guarantee transfer to a new company, region or future climate period.

10 Conclusions and next research steps

10.1 What this project established

The project created a reproducible chain from heterogeneous event records to a national WWTW-linked analytical dataset. It introduced fitted long-duration tail parameters as spatial wastewater signatures, quantified how those parameters co-vary with environmental and system characteristics in one-variable analyses, and then tested whether the complete external information set can predict unseen sites in multivariable supervised regression models.

The final evidence supports three broad conclusions. First, long-duration CSO behaviour contains systematic spatial structure rather than being purely idiosyncratic. Second, the strongest explanatory patterns are distributed across urban-system scale, hydrogeology and wetness/rainfall rather than one single driver. Third, those static and annual-scale variables provide only modest predictive skill, implying that important hydraulic and temporal state information remains missing.

10.2 Priority next steps

The most productive extension is not an unrestricted algorithm search. Priority should be given to event-aligned rainfall and antecedent wetness, groundwater state, network/storage and pumping variables where available, improved beta-lambda reliability modelling, and transfer-aware validation across companies and space. A hierarchical event-level survival model is a natural future endpoint because it can use every long-duration event directly rather than treating fitted beta and lambda as error-free labels.

EVIDENCE NOTE The report should end on the answer, not on the methodology. Keep the final conclusion to one page in the submitted version: what was built, what was found, how strong the evidence is, and what the next decisive dataset would be.

A Data provenance and company coverage

[TO COMPLETE: Insert full source/version/year table, EIR provenance, cleaned-file paths and monitoring-period definitions.]

B Full variable dictionary

[TO COMPLETE: Insert machine-readable field names, readable names, units, source, extraction method, transformation, missingness and role.]

C Complete univariate result atlas

Use the appendix for the full beta, lambda, duration and frequency scatterplots. The main body should retain only the forest plots and a few mechanistically important examples.

D Machine-learning technical appendix

Include complete 70/15/15 split counts, hyperparameter grids, validation tables, bootstrap intervals, original-unit errors, spatial robustness, company sensitivity, permutation importance and learning-curve data. Do not repeat the main-text narrative.

E Reproducibility

Record repository commit, source hashes, Python/scikit-learn versions, random seed, authoritative master paths and figure-generation scripts.

References

References

- [1] T. Giakoumis and N. Voulvoulis. Combined sewer overflows: relating event duration monitoring data to wastewater systems' capacity in England. *Environmental Science: Water Research & Technology*, 9, 707–722, 2023.
- [2] Y. Wang, B. Huang and D. Z. Zhu. Assessment of rainfall-derived inflow and infiltration in sewer systems with machine learning approaches. *Water Science & Technology*, 89(8), 1928–1945, 2024.
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- [4] UCL CSO Spatial Drivers Project. Internal project master context, statistical analysis outputs and machine-learning release, August 2026.